Resilience Academy

SDI Training for ICT Professionals

Linux Basics for ICT Users

10 recipes you can copy-paste — not 100 commands to memorise

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Session Overview

What we will do:

1. Understand *why* Linux matters for SDI
2. Open a terminal window
3. Learn 10 essential commands as **recipes**
4. Practice with two guided exercises

What we will NOT do:

- Become Linux system administrators
- Memorise dozens of commands
- Write scripts or programs

The Cooking Analogy

You do not need to understand food chemistry to follow a recipe. Similarly, you do not need to understand how Linux works internally — you just need to follow **10 simple recipes** (commands) that we will practise together.

Keep this slide deck open as a reference. Every command is written out so you can copy-paste.

Why Linux?

Almost all SDI software runs on Linux:

- GeoNode — Linux
- GeoServer — Linux
- PostgreSQL/PostGIS — Linux
- Docker containers — Linux inside
- CRD production server — Ubuntu Linux

Why not Windows?

- Linux is **free** — no licence costs for servers
- Linux is **stable** — servers run for years without rebooting
- Linux is the **industry standard** for web servers worldwide (over 95% of web

Good News for Windows Users

You do **not** need to replace Windows. There are two easy options:

1. **WSL2** (Windows Subsystem for Linux) — runs a Linux terminal *inside* Windows. No dual-boot needed.
2. **Docker Desktop** — runs Linux containers automatically. You already installed this.

Mac Users

macOS is Unix-based. The same

Opening a Terminal

A **terminal** is simply a text window where you type commands instead of clicking buttons.

Ubuntu Desktop

1. Press `Ctrl` + `Alt` + `T`
   OR search for "Terminal" in the application menu
2. A dark window appears with a blinking cursor
3. You are ready to type commands

Windows 10/11 with WSL2

1. Open the **Start Menu**
2. Search for **"Ubuntu"** (or "WSL")
3. Click to open — a terminal window appears
4. Alternatively: open **Windows Terminal** and select the Ubuntu tab

What You See

```
masoud@laptop:~$
```

Recipes 1–3: Where Am I? What Is Here? Go Somewhere.

Recipe 1: `pwd` — "Where am I right now?"

```
pwd
```

Prints the full path of your current folder.

Example output: `/home/masoud` — you are in your home folder.

Recipe 2: `ls -la` — "What files and folders are here?"

```
ls -la
```

Lists everything in the current folder, including hidden files and details like size and date.

Tip: Use just `ls` for a simpler, shorter listing.

Recipe 3: `cd folder` — "Go to a different folder"

```
cd Documents # go into the Documents folder
```

Recipes 4–6: Create, Copy/Move/Delete, View

Recipe 4: `mkdir name` — "Create a new folder"

```
mkdir sdi_training # creates a folder called sdi_training
mkdir -p data/shapefiles # creates nested folders in one go
```

Like right-clicking and choosing "New Folder" in File Manager.

Recipe 5: `cp`, `mv`, `rm` — "Copy, Move, Delete"

```
cp report.pdf backup_report.pdf # copy a file
mv old_name.txt new_name.txt # rename (or move) a file
rm unwanted_file.txt # permanently delete a file
rm -r old_folder/ # delete a folder and everything in it
```

Warning: `rm` does NOT have a recycle bin. Deleted files are gone forever. Double-check before pressing Enter.

Recipes 7–8: Edit a File, Administrator Mode

Recipe 7: `nano file` — "Edit a text file"

```
nano myfile.txt
```

A simple text editor that works inside the terminal.

Three things to remember inside nano:

1. Type your text normally (the keyboard works as expected)
2. **Save:** Press `Ctrl + O`, then press `Enter`
3. **Exit:** Press `Ctrl + X`

The shortcuts are shown at the bottom of the nano screen. The ^ symbol means "Ctrl".

Recipe 8: `sudo command` — "Run as administrator"

```
sudo apt update # update the list of available software
sudo apt install qgis # install QGIS (requires admin rights)
```

Recipes 9–10: Check Disk Space, Fetch from Web

Recipe 9: `df -h` — "How much disk space do I have?"

```
df -h
```

Shows disk usage in human-readable format (GB, MB). Look for the row with / or `/home` to see your main disk.

Why it matters: Docker images and spatial datasets can be large. If you run out of space, GeoNode will stop working.

Recipe 10: `curl URL` — "Download something from the internet"

```
curl https://crd.resilienceacademy.ac.tz/api/v2/datasets
```

Fetches data from a web address and displays it in the terminal. This is how software talks to web services like CRD behind the scenes.

To save the result to a file instead:

Quick Reference: All 10 Recipes on One Slide

#	Command	What It Does
1	`pwd`	Show current folder path
2	`ls -la`	List files with details
3	`cd folder`	Change to a different folder
4	`mkdir name`	Create a new folder
5	`cp / mv / rm`	Copy, move (rename), or delete files
6	`cat file`	Display file contents
7	`nano file`	Edit a text file (Ctrl+O save, Ctrl+X exit)
8	`sudo command`	Run a command as administrator
9	`df -h`	Check disk space
10	`curl URL`	Fetch data from a web address

You Now Know Enough Linux

These 10 commands cover 90% of what you will need during this training. You can always come back to this slide as a cheat sheet.

Guided Exercise 1: Navigate, Create, View, Delete

Open your terminal and type each command below. Press Enter after each line.

```
# Step 1: Check where you are
pwd

# Step 2: Go to your home folder (just in case)
cd ~

# Step 3: Create a training folder
mkdir sdi_training

# Step 4: Go inside it
cd sdi_training

# Step 5: Create a simple text file
nano hello.txt
# Type: "Hello from SDI Training!" then press Ctrl+O, Enter, Ctrl+X

# Step 6: Verify the file exists
ls -la
```

Guided Exercise 2: Fetch Data from CRD's API

Let us use `curl` to talk to CRD — the same way GeoNode and QGIS do behind the scenes.

```
# Step 1: Ask CRD for a list of datasets (first page)
curl https://crd.resilienceacademy.ac.tz/api/v2/datasets

# You will see a wall of text - that is JSON data (machine-readable)

# Step 2: Save it to a file instead
curl -o crd_datasets.json \
  https://crd.resilienceacademy.ac.tz/api/v2/datasets

# Step 3: Check the file was created
ls -la crd_datasets.json

# Step 4: Look at the first few lines
head -20 crd_datasets.json

# Step 5: Clean up
rm crd_datasets.json
```

Session 3 Summary

What we learned:

- Linux is the standard OS for SDI servers — but you do not need to become an expert
- The **terminal** is just a text-based way to give instructions to your computer
- **10 recipes** cover everything you need:
  - Navigate: `pwd`, `ls`, `cd`
  - Manage files: `mkdir`, `cp/mv/rm`
  - View and edit: `cat`, `nano`
  - Admin and info: `sudo`, `df -h`
  - Fetch from web: `curl`
- CRD's API returns data you can access from the terminal

Tips for Success

- **Copy-paste** is your friend — do not try to memorise commands
- **Tab key** auto-completes file and folder names
- **Up arrow** repeats your last command
- If something goes wrong, `Ctrl + C` cancels the current command

Next Session

Session 4: Environment Setup

We install Docker Desktop and QGIS —